Zaid Ahmad Khan

F-1 STEM OPT eligible · Available full-time May 2027

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SUMMARY

Mechanical design engineer (MS, May 2027) with **3+ years of SolidWorks/Fusion 360** parts, assemblies, GD&T drawings and BOMs, plus hands-on CNC, sheet metal and assembly. Designed an 8-part AHSS enclosure for a Ford Mach-E (160 L of cells, 68 fasteners) that held a **100 kN side-pole load at a minimum safety factor of 1.5** in FEA, and specified die angles that compensate for spring back. Led a 5-member team to design, 3D print and wind-tunnel test a wing, **cutting 25% material** with a dovetail split joint and reaching **90% agreement with theory**. Delivered a 20+ part SolidWorks prototype with 10 drawings at Cupp Golf; built **15 autonomous vehicles** at MIT; waterjet-cut **30+ race-car parts**.

SKILLS

- **CAD & Drafting:** SolidWorks (assemblies, drawings, sheet metal), Fusion 360, AutoCAD, GD&T (ASME Y14.5), BOMs, DFM/DFA.
- **Fabrication:** CNC waterjet, shop machining, sheet metal, 3D printing, soldering/PCB assembly, electromechanical assembly.
- **Analysis & Software:** Ansys Mechanical & Fluent, Fusion 360 FEA, hand calcs, MATLAB, Python, Git, Microsoft Office.

WORK EXPERIENCE

CAD Intern | Cupp Golf, Los Angeles, CA (Remote) *Sep 2026 – Present*

- Modeled a 20+ component Radius Golf Aid prototype assembly in SolidWorks; used mates to check motion and interference.
- Released 10 detail/assembly drawings with manufacturing notes and revision tables, and a BOM of 15+ McMaster-Carr parts.
- Reviewed design safety, structural integrity and commercial viability with the founders over 10+ weeks.

Autonomous RACECAR Teaching Assistant | MIT Beaver Works Summer Institute, Cambridge, MA *Jun 2026 – Aug 2026*

- Built and maintained 15 RACECAR Neo autonomous vehicles: 3D-printed mounts, bolted/magnetic attachments, cable routing.
- Soldered custom power-distribution PCBs, integrated Teensy/Raspberry Pi controls and sensors, and debugged fleet faults.
- Taught PID control and test procedures to high-school students and U.S. Air Force pilots.

CFD Engineering Intern | Flowthermolab Ltd, Nottingham, UK (Remote) *Jan 2025 – Aug 2025*

- Produced 5+ GD&T drawings of a globe valve assembly and prepared its CAD model for Ansys Fluent meshing.
- Simulated flow at valve openings of 1–20 mm, quantifying a ~20% decrease in pressure drop to support valve sizing.

Waterjet Technician | Salford Racing (Formula Student), Salford, UK *Jan 2023 – Jul 2023*

- Waterjet-cut 30+ metal and carbon-fiber chassis, exhaust and structural parts to drawing and H&S procedures.
- Prepared 20+ part cut files in Illustrator; built 2 shadow boards, improving workshop efficiency by 15%.

ENGINEERING PROJECTS

EV Battery Enclosure Design | Build Fellowship / Open Avenues | *Fusion 360, FEA, GD&T, BOM* *Feb 2026 – Apr 2026*

- Designed an 8-component AHSS sheet-metal enclosure (2062 × 1343 × 128 mm) with cooling plates and 68 M6 fasteners.
- FEA of 100 kN side-pole and 19.6 kN underfloor loads (FMVSS 305 / ECE R100), min. safety factor 1.5; verified by hand calcs.
- Specified roll forming, an 81.28° die angle for spring back (K = 0.984), and multi-pulse spot welds to limit LME.

Transonic Wing Prototyping & Wind Tunnel Testing | *SolidWorks, 3D printing, XFOIL* *Jan 2024 – Apr 2024*

- Led a 5-member team to design a 0.5 m span lofted wing in SolidWorks with 10 pressure taps built into the geometry.
- Split the wing with a dovetail joint to fit the printer, cutting material 25% through prototype iteration.
- Tested at 5–25 m/s, located boundary-layer separation, and reached 90% agreement with theory/XFOIL.

Screw Gauge Manufacturing | Trafford College, Manchester, UK *Oct 2023*

- Machined a mild-steel screw gauge from GD&T blueprints using hand tools and shop machinery.

EDUCATION

Illinois Institute of Technology, Chicago, IL | MS, Mechanical & Aerospace Engineering | GPA 3.8/4.0 *May 2027*

- Commencement Honor Marshal; 40% merit-based tuition scholarship.

University of Salford, Salford, UK | BEng (Hons) Aeronautical Engineering, First Class Honours *Jun 2024*

- Ranked 2nd in the program; awarded the Gold Excellence Scholarship.

LEADERSHIP

Technical Project Director | AIAA at Illinois Tech, Chicago, IL *Sep 2025 – Present*

- Run weekly Fusion 360 CAD workshops for 30+ members and lead 10+ executive board meetings.

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AHSS EV Battery Enclosure | *Build Fellowship Hub · Fusion 360, FEA, GD&T, BOM, DFM* *Feb 2026 – Apr 2026*

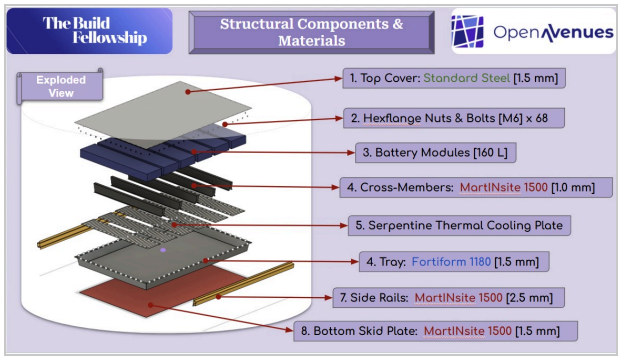
Designed a manufacturable, serviceable steel enclosure for a 160 L Ford Mach-E battery pack during an 8-week consulting program. It was validated against FMVSS 305 / ECE R100 side-impact and underfloor-intrusion targets and documented with drawings and a BOM.

- **8-part sheet-metal assembly:** CR5 top cover (2.5 mm), Fortiform 1180 housing tray (1.5 mm), 5 MartINsite 1500 cross-members (1.0 mm), 2 roll-formed side rails (2.5 mm), a bottom skid plate (1.5 mm) and 6 Al 6061 serpentine cooling plates.
- **Fastening:** 68 ISO 4162 M6 × 12 class 10.9 hex-flange bolts with ISO 4161 nuts. Joint sized by hand calculation (preload ≈ 81% of yield, 18 kN joint yield limit).
- **Documentation:** 8-line BOM (152 components) with material and gauge, plus dimensioned drawings of the side rail, cross-member and tray using section and break-out views.
- **FEA:** three load cases: 3 g distributed top load (SF > 15), **100 kN side-pole load (min. SF ≈ 1.5)**, and 150 mm bottom intrusion at 19.6 kN (SF > 15).
- **Section properties:** I = 557k mm^4 (cross-member) and 685k mm^4 (side rail), used to check bending and buckling resistance.
- **Forming:** roll forming specified for the side rails (R/t of 3–4 for AHSS above 1000 MPa), and an **81.28° die angle** to hold the 80° tray bend after springback (K_s = 0.984).
- **Joining & corrosion:** EG zinc coating and multi-pulse resistance spot welds to limit liquid-metal embrittlement in the martensitic cross-members.

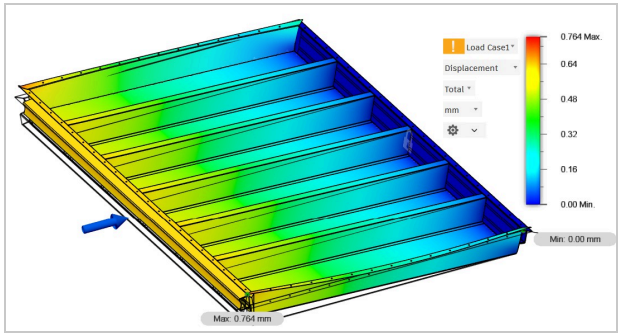


Item	Qty	Part Name	Material
1	1	Top Cover [2.5 mm]	CR5
2	68	Hex-Flange Bolts [ISO 4162 M6 x 12]	Steel 10.9
3	68	Hex-Flange Nuts [ISO 4161 M6]	Steel 10
4	5	Cross-Member Beams [1.0 mm]	MartINsite 1500
5	2	Side Rails [2.5 mm]	MartINsite 1500
6	6	Thermal Plates	Al 6061
7	1	Bottom Plate [1.5 mm]	MartINsite 1500
8	1	Main Housing Tray [1.5 mm]	Fortiform 1180

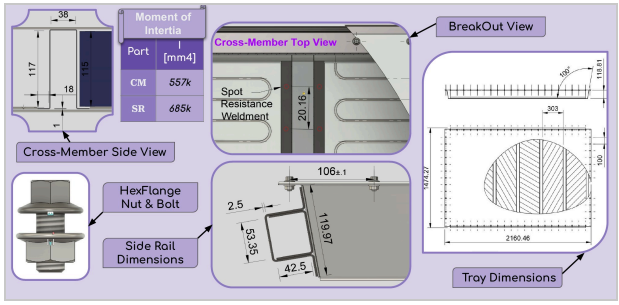
Bill of materials: 8 line items, 152 components



Exploded view of the 8-part enclosure assembly



Static stress: 100 kN side-pole load



Cross-member, side-rail & tray dimensions

Radius Golf Aid Prototype | *CAD Intern, Cupp Golf · SolidWorks, BOM, Revision Control* *Sep 2026 – Present*

Taking a new golf training aid from concept to a released prototype design, working directly with the company's founders (remote).

- Modeled and assembled a **20+ component prototype** in SolidWorks, using mates to verify range of motion and check for interference.
- Released **10 detail and sub-assembly drawings** with dimensions, manufacturing and application notes, and revision tables.
- Built the assembly **BOM**, sourcing 15+ purchased components from McMaster-Carr; presented design reviews on safety, structural integrity and commercial viability.

Models and drawings are proprietary to Cupp Golf and are not shown.

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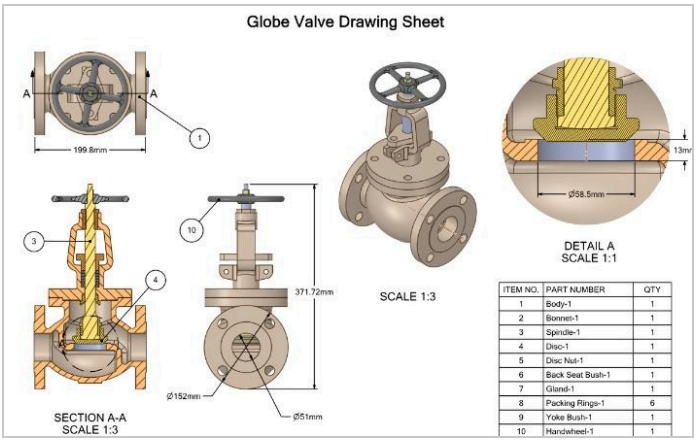
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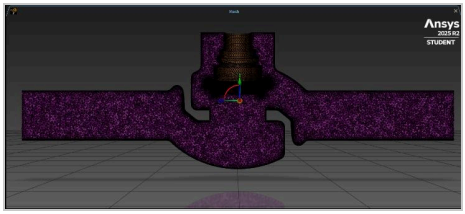
Globe Valve Assembly & Flow Analysis | *CFD Intern, Flowthermolab · SolidWorks, Ansys Fluent* *Jan 2025 – Aug 2025*

Modeled a flanged globe valve, documented it to drawing standards, and quantified how valve opening affects pressure drop.

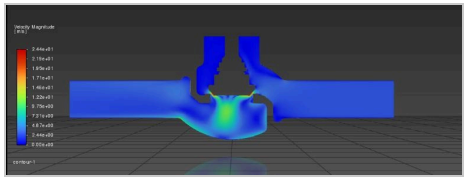
- Modeled a **10-part assembly** in SolidWorks: body, bonnet, spindle, disc, disc nut, back-seat bush, gland, packing rings, yoke bush and handwheel.
- Produced **5+ drawing sheets** with GD&T, Section A-A (1:3), Detail A (1:1), overall dimensions and a ballooned parts list.
- Extracted and cleaned the internal fluid domain from the solid model so it could be meshed in Ansys Fluent.
- Simulated openings of 1, 5 and 20 mm: pressure drop **fell ~20%** as the opening increased. Contours located jetting, mixing and cavitation risk at the disc and seat.



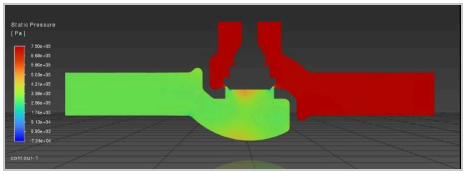
Drawing sheet with section, detail & parts list



Internal fluid-domain mesh



Velocity contour, 5 mm opening



Static pressure, 5 mm opening

Waterjet Fabrication | *Waterjet Technician, Salford Racing (Formula Student) · CNC Waterjet* *Jan 2023 – Jul 2023*

Manufactured structural components for the Formula Student car to drawing and improved machine-shop workflow.

- Cut **30+ metal and carbon-fiber parts** for the chassis, exhaust and structural assemblies on an industrial waterjet, to engineering drawings and H&S procedures.
- Converted **20+ part drawings** into machine-ready 2D cut files in Adobe Illustrator, and checked finished parts against drawing tolerances for fit-up.
- Designed and cut **2 tool shadow boards**, improving workshop workflow efficiency by 15%.





Waterjet cutting & shadow board

Machining & Structures Labs | *University of Salford · Trafford College*

- Machined a **mild-steel screw gauge** from GD&T blueprints using hand tools and shop machinery (Trafford College, Oct 2023).
- Experimental testing of **beam bending, buckling, tensile fracture and hardness**, comparing measured results with hand calculations.

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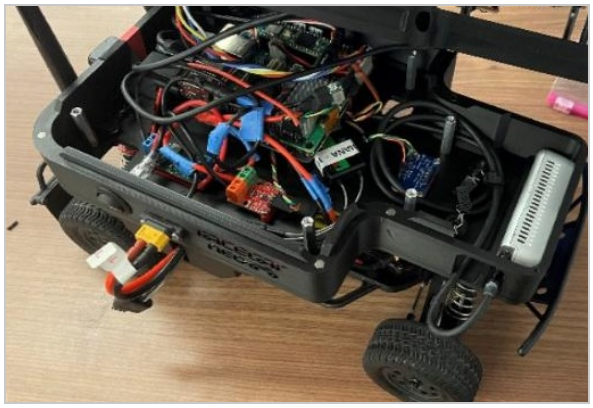
Autonomous RACECAR Fleet | *Teaching Assistant, MIT BWSI · Electromechanical Assembly* *Jun 2026 – Aug 2026*

Upgraded, fabricated and maintained a fleet of autonomous robots for student engineering cohorts in a lab environment.

- Built and field-tested **15 RACECAR Neo units** integrating LiDAR, a camera, a Raspberry Pi 5, a Teensy microcontroller, an ESC and Ackermann steering.
- **3D printed structural mounts** and laid out bolted, screwed and magnetic attachment points for sensors, compute and batteries.
- Fabricated **custom power-distribution PCBs** using solder paste and iron, and integrated a Teensy to divert power away from the Raspberry Pi 5.
- Built wire harnesses with crimped connectors and heat-shrink; diagnosed electrical, firmware and Pi–Teensy communication faults across the fleet.
- Taught PID control and test procedures to high-school students and U.S. Air Force pilots, and designed track layouts to stress-test durability.



Upgraded RACECAR Neo



Internal components & cable routing

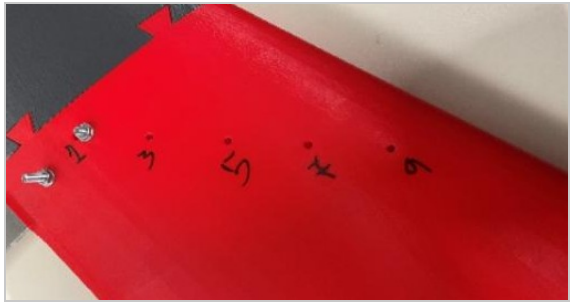
Transonic Wing Prototyping & Wind Tunnel Validation | *Team Lead · SolidWorks, FDM, XFOIL* *Jan 2024 – Apr 2024*

Led a 5-member team to design, 3D print and wind-tunnel test an instrumented wing, then validated measured pressures against theory.

- Modeled a 0.5 m span, 0.2 m chord wing in SolidWorks by **lofting two transonic airfoils** between root and tip.
- Built **10 pressure-tap channels** into the CAD geometry across the upper and lower surfaces, within the printer's resolution limits.
- Split the wing with a **dovetail joint** to fit the printer build volume; iterated a first prototype to check assembly tolerances and **cut material by 25%** by tuning PLA infill and wall thickness.
- Tested at **5–25 m/s** on a three-strut mount, located boundary-layer separation, and converted tap pressures to lift, reaching **90% agreement** with theory/XFOIL.



Wing on three-strut mount in test section



Pressure-tap locations

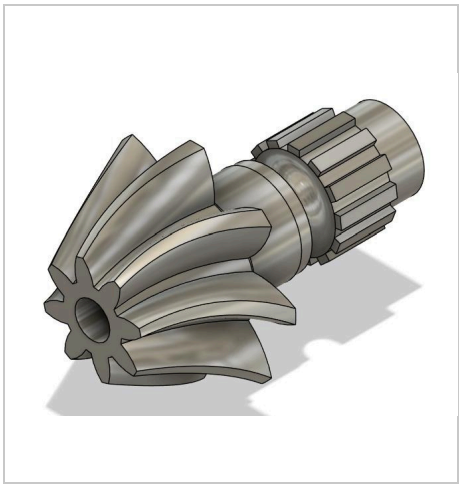
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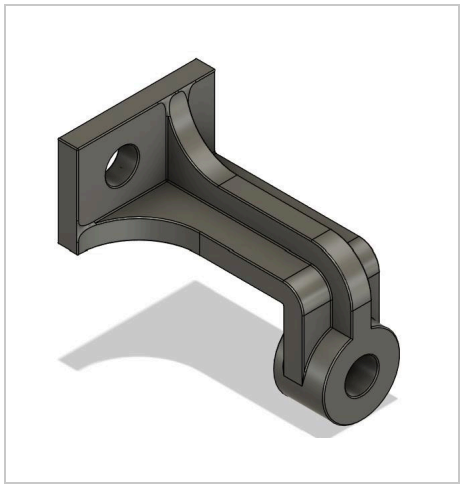
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CAD Models | *Autodesk Fusion 360*



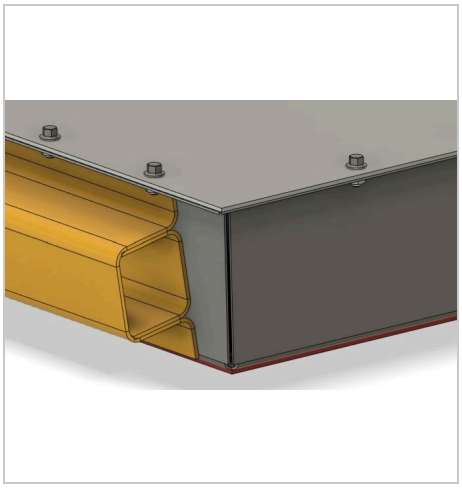
Helical Bevel Pinion

Helical bevel teeth with an integral splined shaft.



Cantilevered Mounting Bracket

Bolted base flange, fillet stiffening and a pin bore.



EV Enclosure Corner Detail

Roll-formed 2.5 mm side rail under a bolted M6 cover.

Elliptical Wing CAD Workshops | *Technical Project Director, AIAA at Illinois Tech · Fusion 360* *Sep 2025 – Present*

Run weekly parametric-CAD workshops that prepare 30+ members for the AIAA Design/Build/Fly competition.

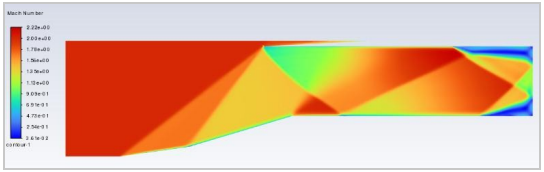
- Modeled a complete **elliptical wing structure** (spars, ribs, struts, flaps and slats) as a teaching assembly.
- Modeled and assembled a surface gauge and a globe valve in SolidWorks as drafting exercises.
- Lead 10+ executive board meetings on project scope, timelines and budget.



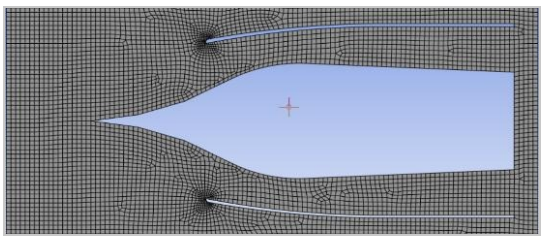
Elliptical wing structure

Supersonic Ramjet Inlet CFD | *Research Project · Ansys Fluent, DesignModeler*

- Modeled **2D axisymmetric inlet geometries** for Mach 2, 2.5 and 3 to compare compression performance.
- Used a density-based solver with the Spalart-Allmaras model to analyze shock interaction and boundary-layer growth.
- Ran a mesh independence study up to **255,420 elements**, converging residuals to 10^{-3}.



Mach contour, inlet A1 at Mach 2



Structured mesh around the inlet centerbody